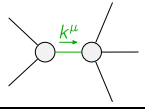
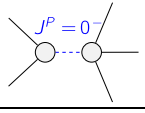
Abbreviations used in matrices

$$Y_1 == v + 2\,\sigma_1 \;\&\&\; Y_2 == v - \sigma_2 \;\&\&\; Y_3 == 2\,\sigma_1 + \sigma_2 \;\&\&\; \text{Det}\,(0^+) == -\frac{1}{2}\,(1 + 2\,k^2)\,M_P^2\,\upsilon_2 \;\&\&$$

$$\text{Det}\,(1^+) == -\frac{2}{3}\,k^2\,(1 + k^2)\,(v + 2\,\sigma_1)\,\sigma_2\,\lambda \;\&\&\; \text{Det}\,(2^+) == -k^2\,(2\,\sigma_1 + \sigma_2)$$

Added source term(s):	$f^{\alpha\beta}\ \tau_{\alpha\beta} + \omega^{\alpha\beta\chi}\ \sigma_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\tau_{0^+}^{\#2} == 0$	1	$\partial_\beta\partial_\alpha\tau^{\alpha\beta} == 0$
$\tau_{0^+}^{\#1} - 2\,i\,k\,\sigma_{0^+}^{\#1} == 0$	1	$\partial_\beta\partial_\alpha\tau^{\alpha\beta} == \partial_\beta\partial^\beta\tau^\alpha{}_\alpha + 2\,\partial_\chi\partial^\chi\partial_\beta\sigma^{\alpha\beta}{}_\alpha$
$\tau_{1^-}^{\#2\,\alpha} == 0$	3	$\partial_\chi\partial_\beta\partial^\alpha\tau^{\beta\chi} == \partial_\chi\partial^\chi\partial_\beta\tau^{\alpha\beta}$
$\tau_{1^-}^{\#1\,\alpha} + 2\,i\,k\ \sigma_{1^-}^{\#2\,\alpha} == 0$	3	$\partial_\chi\partial_\beta\partial^\alpha\tau^{\beta\chi} == \partial_\chi\partial^\chi\partial_\beta\tau^{\beta\alpha} + 2\,\partial_\delta\partial^\delta\partial_\chi\partial_\beta\sigma^{\alpha\beta\chi}$
$\tau_{1^+}^{\#1\,\alpha\beta} - i\,k\ \sigma_{1^+}^{\#2\,\alpha\beta} == 0$	3	$\partial_\chi\partial^\alpha\tau^{\beta\chi} + \partial_\chi\partial^\beta\tau^{\chi\alpha} + \partial_\chi\partial^\chi\tau^{\alpha\beta} + 2\,\partial_\delta\partial_\chi\partial^\beta\sigma^{\alpha\chi\delta} == \partial_\chi\partial^\alpha\tau^{\chi\beta} + \partial_\chi\partial^\beta\tau^{\alpha\chi} + \partial_\chi\partial^\chi\tau^{\beta\alpha} + 2\,\partial_\delta\partial_\chi\partial^\alpha\sigma^{\beta\chi\delta} + 2\,\partial_\delta\partial^\delta\partial_\chi\sigma^{\alpha\beta\chi}$
$\sigma_{2^-}^{\#1\,\alpha\beta\chi} == 0$	5	$3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\sigma^{\beta\delta\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\sigma^{\beta\delta}{}_\delta + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\sigma^{\alpha\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\sigma^{\alpha\delta\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\sigma^{\chi\delta\alpha} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\sigma^{\alpha\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\sigma^{\alpha\delta\beta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\sigma^{\beta\chi\alpha} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\sigma^{\delta\epsilon}{}_\delta + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\sigma^{\beta\delta\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\sigma^{\alpha\delta}{}_\delta == 3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\sigma^{\alpha\delta\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\sigma^{\alpha\delta}{}_\delta + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\sigma^{\beta\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\sigma^{\beta\delta\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\sigma^{\chi\delta\beta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\sigma^{\beta\delta\alpha} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\sigma^{\alpha\beta\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\sigma^{\alpha\chi\beta} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\sigma^{\delta\epsilon}{}_\delta + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\sigma^{\alpha\delta\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\sigma^{\beta\delta}{}_\delta$
$\tau_{2^+}^{\#1\,\alpha\beta} == 0$	5	$4\,\partial_\delta\partial_\chi\partial^\beta\partial^\alpha\tau^{\chi\delta} + 2\,\partial_\delta\partial^\delta\partial^\beta\partial^\alpha\tau^\chi{}_\chi + 3\,\partial_\delta\partial^\delta\partial_\chi\partial^\chi\tau^{\alpha\beta} + 3\,\partial_\delta\partial^\delta\partial_\chi\partial^\chi\tau^{\beta\alpha} + 2\,\eta^{\alpha\beta}\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial_\chi\tau^{\chi\delta} == 3\,\partial_\delta\partial^\delta\partial_\chi\partial^\alpha\tau^{\beta\chi} + 3\,\partial_\delta\partial^\delta\partial_\chi\partial^\alpha\tau^{\chi\beta} + 3\,\partial_\delta\partial^\delta\partial_\chi\partial^\beta\tau^{\alpha\chi} + 3\,\partial_\delta\partial^\delta\partial_\chi\partial^\beta\tau^{\chi\alpha} + 2\,\eta^{\alpha\beta}\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\tau^\chi{}_\chi$
Total # constraint(s):	21	

Resolved pole(s)	# d(s).o.f	Square mass	Residue
	2	0	$\dfrac{\frac{5+4\,p^2}{v+2\,\sigma_1} + \frac{4\,(1+2\,p^2)}{-v+\sigma_2} - \frac{1+4\,p^2}{2\,\sigma_1+\sigma_2}}{p^2}$
	1	$\dfrac{\sigma_2\,\lambda}{\sigma_2}$	$-\dfrac{1}{2\,\sigma_2}$
Resolved unitarity condition(s):	$(\sigma_1 \le 0 \;\&\&\; \sigma_2 < 0 \;\&\&\; ((v < \sigma_2 \;\&\&\; \sigma_2\,\lambda < 0) \,||\, (v > -2\,\sigma_1 \;\&\&\; \sigma_2\,\lambda < 0))) \,||\, (\sigma_1 > 0 \;\&\&$ $((\sigma_2 < -2\,\sigma_1 \;\&\&\; ((v < \sigma_2 \;\&\&\; \sigma_2\,\lambda < 0) \,||\, (v > -2\,\sigma_1 \;\&\&\; \sigma_2\,\lambda < 0))) \,||\, (-2\,\sigma_1 < \sigma_2 < 0 \;\&\&\; -2\,\sigma_1 < v < \sigma_2 \;\&\&\; \sigma_2\,\lambda < 0)))$		